

Companion LIV: The Quadratic Lensing Estimator in the Relaxation-Driven Cyclic Cosmology

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Abstract

Quadratic estimators provide the standard method for reconstructing the CMB lensing potential from mode coupling in temperature and polarization fields. In the standard cosmological model, the estimator response and reconstruction noise are determined by the lensing potential power spectrum and the statistical properties of the CMB. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the lensing kernel, the mode-coupling structure, and the estimator normalization in a characteristic way. This Companion develops a continuous description of the quadratic lensing estimator in RDCC, deriving how the relaxation scale influences the estimator response, the reconstruction noise, and the coherence of the reconstructed potential. The resulting framework provides a distinctive observational signature of the RDCC gravitational field.

1 Introduction

CMB lensing reconstruction relies on the fact that gravitational lensing couples modes of the CMB temperature and polarization fields. Quadratic estimators exploit this coupling to reconstruct the lensing potential ϕ . The estimator response and reconstruction noise depend on the lensing potential power spectrum and the statistical properties of the CMB.

In the standard cosmological model, the lensing potential evolves smoothly and scale-independently. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the lensing potential, the mode-coupling structure, and the estimator response in a scale-dependent manner, producing a distinctive signature in the reconstructed lensing potential.

2 The Quadratic Lensing Estimator

The standard quadratic estimator is given by

$$\hat{\phi}_{LM} = N_L \sum_{\ell m} \sum_{\ell' m'} X_{\ell m} X_{\ell' m'} \mathcal{W}_{\ell \ell' L}^{m m' M}, \quad (1)$$

where X denotes temperature or polarization fields, $\mathcal{W}$ is a weighting kernel, and N_L is the normalization.

The estimator response is defined as

$$\langle \hat{\phi}_{LM} \rangle = R_L \phi_{LM}. \quad (2)$$

In the standard cosmological model, $R_L \approx 1$ on most scales.

3 The RDCC Modification of the Estimator Response

In RDCC, the lensing potential evolves as

$$\phi_{\text{RDCC}} = \phi \left[1 - \chi_\phi \left(\frac{L}{L_{\text{rel}}} \right)^\alpha \right]. \quad (3)$$

The estimator response becomes

$$R_{L,\text{RDCC}} = R_L \left[1 - \chi_R \left(\frac{L}{L_{\text{rel}}} \right)^\alpha \right]. \quad (4)$$

This modification reflects the scale-dependent evolution of the gravitational potential.

4 Reconstruction Noise in RDCC

The reconstruction noise is given by

$$N_L^{(0)} = \left[\sum_{\ell\ell'} \frac{|\mathcal{W}_{\ell\ell'L}|^2}{C_\ell^{XX} C_{\ell'}^{XX}} \right]^{-1}. \quad (5)$$

In RDCC, the CMB power spectra become

$$C_{\ell,\text{RDCC}}^{XX} = C_\ell^{XX} \left[1 - \chi_X \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (6)$$

The reconstruction noise becomes

$$N_{L,\text{RDCC}}^{(0)} = N_L^{(0)} \left[1 + \chi_N \left(\frac{L}{L_{\text{rel}}} \right)^\alpha \right]. \quad (7)$$

This modification increases noise at small scales and decreases noise at large scales.

5 Coherence of the Reconstructed Potential

The reconstructed potential is

$$\hat{\phi}_{LM,\text{RDCC}} = \hat{\phi}_{LM} \left[1 - \chi_{\text{rec}} \left(\frac{L}{L_{\text{rel}}} \right)^\alpha \right]. \quad (8)$$

The relaxation mechanism enhances the coherence of large-scale modes. The reconstructed potential therefore exhibits:

- enhanced large-scale structure,
- suppressed small-scale fluctuations,
- a characteristic turnover near L_{rel} ,
- a modified estimator response function.

These signatures provide a direct observational probe of the RDCC gravitational field.

6 Conclusion

The quadratic lensing estimator in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the gravitational potential. The relaxation mechanism modifies the estimator response, the reconstruction noise, and the coherence of the reconstructed potential in a scale-dependent manner, producing distinctive signatures in CMB lensing reconstruction. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of CMB lensing, potential reconstruction, and the interplay between matter and gravity in RDCC.

References

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